

Zhengkao Xu

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Education

University of Science and Technology Beijing (USTB) — B.Eng. candidate, Computer Science and Technology Sep. 2024 – Jun. 2028 (expected)

School of Computer and Communication Engineering. GPA 3.81/4.00, weighted average 89.6/100, major rank 10/137.

Carnegie Mellon University — Collaborative course, Algorithms for Big Data (Prof. David P. Woodruff) Oct. 11, 2024 – Jan. 5, 2025

Grade: 88.70/100.

Westlake University — 2026 PEBBLE BioFusion Workshop, Center for Interdisciplinary Studies Jul. 24 – Aug. 4, 2026

Research Experience

GRACE: Graph-Aware Routing Along with Compression for End-to-End Optimization

Fourth author · Sep. 2025 – Present

- Implemented the PPO training, evaluation, and logging pipeline for joint routing and in-network compression with GNN-encoded network states.
- Designed cross-topology generalization experiments by evaluating a policy trained on one topology directly on unseen topologies, without additional training.
- Ran workload, parameter-sensitivity, baseline, and ablation experiments for the current IEEE TNSM Major Revision.

RobustVidBench: Benchmarking Robustness of Multimodal Video Understanding

Fourth author · Sep. 2025 – Present

- Worked on video preprocessing for the benchmark, including download, clipping, transcoding, sampling, format normalization, audio handling, and metadata cleaning.
- Implemented controlled degradations: bitrate/resolution/frame-rate reduction, frame dropping, stuttering/repeated frames, and audio–video misalignment.
- Participated in QA verification, model evaluation, experimental analysis, and manuscript preparation.

Automated Video Understanding Dataset Construction

Fourth author · Feb. 2026 – Present

- Participated in the design of a Video → Knowledge Graph → Summary → QA data-construction pipeline.
- Implemented core video-processing and QA-generation components and used vLLM for batched inference.
- The project covers 600+ videos; I directly worked on about 150 videos and generated or processed nearly 10,000 QA samples across the main pipeline and experimental variants.

A Recurrent Rab8a-associated Hypoxic Niche in Spatial CRISPR Tumour Data

First author · 2026 PEBBLE BioFusion Workshop, Westlake University · Jul. 2026 – Aug. 2026

- Developed a dual-view representation that combines local cellular-neighborhood information and gene-expression features with graph neural networks.
- Contributed to model implementation, experiments, and manuscript preparation. The work received the Excellent Research Report Award.

Computation–Routing Joint Optimization in UAV Networks

Second author · 2025 – Present

- Led problem formulation, reinforcement-learning implementation, and simulation experiments for joint routing, compute placement, and compression decisions in a shared wireless environment.

Selected Research Outputs

- Xiaolong Cui, Xuebin Tang, Zixu Wang, Zhenghao Xu, Wei Huangfu. “GRACE: Graph-Aware Routing Along with Compression for End-to-End Optimization.” IEEE Transactions on Network and Service Management (TNSM), Major Revision.
- Dongyan Zhang, Peijie Wu, Jingyu Wu, Zhenghao Xu, Xiaotian Hu, Yuqing Zhang. “RobustVidBench: Benchmarking Robustness of Multimodal Video Understanding.” Manuscript in preparation.
- Zhenghao Xu et al. “A Recurrent Rab8a-associated Hypoxic Niche in Spatial CRISPR Tumour Data.” Research manuscript, 2026 PEBBLE BioFusion Workshop, Westlake University.

Patent

- Xiaolong Cui, Xuebin Tang, Zixu Wang, Wei Huangfu, Zhenghao Xu. “A Deep Reinforcement Learning Method and System for Joint Optimization of Computing-Network Routing and In-Network Data Processing.” Chinese Invention Patent Application No. 2026102073230, 2026.

Awards

- Meritorious Winner, Interdisciplinary Contest in Modeling (ICM 2026), Problem E. Team captain; mainly responsible for modeling and implementation.
- Excellent Research Report Award, 2026 PEBBLE BioFusion Workshop, Westlake University.
- First Prize (Beijing), Jingcai Dachuang Innovation and Entrepreneurship Competition.
- Second Prize (Beijing), Challenge Cup Capital Entrepreneurship Plan Competition, “Qingzhi Future” track (team captain).
- Second Prize (Beijing), Challenge Cup Capital Entrepreneurship Plan Competition, “Qingli Grassroots” track.
- First Prize, USTB Cradle Cup Innovation and Entrepreneurship Competition.

Skills

Programming: Python, C/C++, Java, SQL

ML & AI: PyTorch, vLLM, Hugging Face, TensorBoard

Development: Linux, Git, Docker, LaTeX

Scientific computing: NumPy, Pandas, NetworkX, MATLAB

Activities

Class Representative, Computer Science Class 242. Head of Academic Affairs, Suosi Science and Technology Association.